# Minimal jet orders in a positive behavioral SIR model

Tatsuhiko Takimoto
Independent researcher

Version 1.0, 10 November 2026

**Abstract**

For the behavioral SIR model with transmission rate $\beta SI/(1+\alpha I)^k$, we determine the least order of an exact output jet at one time that identifies the rates and hidden initial states, on the positive real domain with known total population. For each fixed, known $k \in \{2, 3, 5/2\}$, generic local identification requires order four, whereas generic identification against all positive competitors requires order five, in the sense of identification on an open dense set of true points. Fourth-order ambiguity is established on a nonempty open set, without a density claim. The fifth-order upper bound is an application of the observation-field theorem of Demin, Pogudin and Rackauckas and a prior identifiability result for this model. To prove that four derivatives do not suffice globally, we give exact certificates for two distinct positive points with the same fourth-order jet and nonzero jet Jacobians at both points. The inverse function theorem extends this ambiguity to a nonempty open set. The certificates have rational isolating intervals and can be checked using Python's standard library. The result concerns exact derivatives, not estimation from noisy samples.

## 1 Model and the least-order question

Consider

$$S' = -\frac{\beta SI}{(1+\alpha I)^k}, \qquad I' = \frac{\beta SI}{(1+\alpha I)^k} - \gamma I, \qquad R' = \gamma I, \qquad y = I. \tag{1}$$

Fix known $k$ and $N_0 > 0$. The unknown initial point lies in

$$\Omega_{N_0} = \{(\alpha, \beta, \gamma, S_0, I_0) \in \mathbb{R}^5_{>0} : S_0 + I_0 < N_0\}, \qquad R_0 = N_0 - S_0 - I_0.$$

All powers use the positive real branch. Solutions are analytic near zero. Since the system is autonomous, the observation time can be translated to zero. Write

$$J_m(\theta) = (I(0), I'(0), \ldots, I^{(m)}(0)). \tag{2}$$

Order $m$ means $m+1$ observed numbers, including the zeroth derivative.

**Definition 1.** *The jet $J_m$ identifies a point $\theta$ locally if there is a neighborhood $V$ of $\theta$ in $\Omega_{N_0}$ such that $J_m^{-1}(J_m(\theta)) \cap V = \{\theta\}$. It identifies $\theta$ globally if this equality holds with $V = \Omega_{N_0}$. Generic local or global identifiability means that the corresponding property holds at every point of an open dense subset of $\Omega_{N_0}$ in the Euclidean topology. In the global definition all competing points have the same known $k, N_0$, and are not restricted to the generic subset.*

**Theorem 2.** *For each fixed, known $k \in \{2, 3, 5/2\}$ and each known $N_0 > 0$, the least jet order for generic local identifiability of $\alpha, \beta, \gamma, S_0, I_0, R_0$ in (1) is four. The least jet order for generic global identifiability against all positive competitors is five: $J_4$ contains five observed numbers, whereas $J_5$ contains six. Moreover, for each of these exponents there is a nonempty open subset of $\Omega_{N_0}$ whose points have distinct positive competitors with the same $J_4$.*

Here $R_0 = N_0 - S_0 - I_0$ is a dependent quantity, so the unknown parameter-state domain has dimension five.

Pant, Saucedo and Pogudin studied this behavioral model and its identifiability in [1, 2, Section 3.6 and Supplement S9]. Their supplement provides the auxiliary state $Z = (1+\alpha I)^k/\beta$. A companion

proof [4] establishes generic identifiability from the complete analytic output for every fixed known real $k \geq 2$. This does not specify the least finite jet order.

The observation-field results of Demin, Pogudin and Rackauckas [3, Theorem 4.12 and Remark 4.18] supply the upper bound used here. Finite-jet reduction and local rank methods are established tools; related identifiability theory appears in [6, 5]. The contribution of this note is the positive fourth-order ambiguity at two regular points for three exponents, which makes the known upper bound sharp. An input–output equation of order two still contains unknown coefficients, so its order is not the number of derivatives needed to recover those coefficients and the initial state at one time.

**Assurance.** The proof combines cited mathematical results, the inverse function and constant rank theorems, and finite exact certificate checks. Separately written checks support the certificate identities and their positivity. This is a prose and computer-assisted proof; it has not been formalized in a proof assistant or reviewed by human experts. The verification scope and software comparisons are recorded in `VERIFICATION-RECORD.md`.

## 2 The fifth-order upper bound

Write $k = p/q$ in lowest terms. The positive coordinate $z = (1 + \alpha I)^{1/q}$ gives the rational two-state model

$$S' = -\frac{\beta S(z^q - 1)}{\alpha z^p}, \qquad z' = \frac{z^q - 1}{q z^{q-1}} \left( \frac{\beta S}{z^p} - \gamma \right), \qquad y = \frac{z^q - 1}{\alpha}. \tag{3}$$

For $q = 1$ ($k = 2, 3$), this is the original model in the coordinate $z = 1 + \alpha I$; no root is taken. It has two states and three unknown constant parameters. Over $\mathbb{C}$, let $\mathcal{O}$ denote its observation field, generated by the output and all its Lie derivatives. Apply [3, Theorem 4.12 and Remark 4.18] with ground subfield $\mathcal{E} = \mathbb{C}$ and no external inputs. The states-plus-parameters bound is derivative order five, or the six observed numbers in $J_5$, hence

$$\mathcal{O} = \mathbb{C}(y, Dy, \ldots, D^5 y). \tag{4}$$

Unknown parameter functions are not supplied to $\mathcal{E}$ as if they were already observed.

For completeness, the bound in this no-input setting also follows by a classical differential-field argument. Put $L = \mathbb{C}(\alpha, \beta, \gamma, S, z)$ and $F_j = \mathbb{C}(y, Dy, \ldots, D^j y)$. The output $y$ is transcendental over $\mathbb{C}$, and $D(F_j) \subset F_{j+1}$. Let $r \geq 1$ be the first index for which $D^r y$ is algebraic over $F_{r-1}$. Such an $r \leq 5$ exists because $\operatorname{trdeg}(L/\mathbb{C}) = 5$. If $G(T) \in F_{r-1}[T]$ is its minimal polynomial, differentiating $G(D^r y) = 0$ expresses $D^{r+1} y$ in $F_r$: the derivatives of the coefficients lie in $F_r$, and the separant $G'(D^r y) \neq 0$ in characteristic zero. Hence $D(F_r) \subset F_r$ and all later derivatives lie in $F_r$. Thus $\mathcal{O} = F_r = F_5$, proving (4) without an additional finite-generation hypothesis. This explains the cited bound in the present setting; it is not a new upper-bound result.

The following is the precise cover statement needed here.

**Lemma 3** (Observable quantities on the cover)**.** *For $(p, q) = (2, 1), (3, 1), (5, 2)$, equip $L = K(S, z)$, $K = \mathbb{C}(\alpha, \beta, \gamma)$, with the derivation in* (3)*, where the five coordinates are algebraically independent and $D|_K = 0$. Its rational constant field is exactly* $\ker D = K$*, and $\alpha, \gamma, \beta^q, S, y$ belong to its observation field $\mathcal{O}$ over $\mathbb{C}$.*

The constant-field assertion is the absence of rational first integrals beyond the parameters, required by [5, Theorem 4.7]. In the fixed companion version, Lemma 3 (Section 2) checks that hypothesis; Lemma 4 and Proposition 5 (Section 3) give the minimal normalized input–output equation; Proposition 6 (Section 4), equations (11)–(13), recovers $\alpha, \gamma, \beta^q$ from its coefficients. Section 5, equation (16), recovers $S$. Theorem 2 of that work is the resulting positive-domain statement; that statement alone is not being used as a substitute for membership in $\mathcal{O}$. Appendix B proves the lemma here, including direct Wronskian recovery of the coefficients in $\mathcal{O}$. Thus the present proof does not need to infer observability from input–output identifiability via [5, Corollary 5.7]. Consequently

$$\alpha, \gamma, \beta^q, S, y \ \in \mathcal{O}. \tag{5}$$

In particular, for $q = 1$ the coefficient formulas recover $\beta$; for $q = 2$ they recover $\beta^2$, not a choice between its complex square roots. This distinction is retained when using (4). Positivity is essential to the subsequent global conclusion; no such conclusion for arbitrary complex or signed parameters is asserted.

**Proposition 4.** *For the three exponents of Theorem 2, $J_5$ identifies a generic true point against every positive competitor at the same $k, N_0$.*

*Proof.* For each of the five quantities $h$ in (5), choose a rational representation $h = A_h(J_5)/B_h(J_5)$ provided by (4). The pullback of each denominator is nonzero as a rational function. Let $B$ be their product, and restrict the true point to $B(J_5) \neq 0$. After clearing model denominators, the equalities $hB_h(J_5) = A_h(J_5)$ are rational identities on the cover.

A positive competitor with the same $J_5$ has exactly the same nonzero denominator values. The model denominators are also nonzero there: $\alpha > 0$ and $z > 1$. Thus the same identities recover $\alpha, \gamma, \beta^q, S_0, I_0$ at the competitor. Positivity gives a unique $q$th root $\beta$, and $R_0 = N_0 - S_0 - I_0$ then agrees.

The positive cover with $S_0 + (z_0^q - 1)/\alpha < N_0$ is a nonempty real open subset of the five-dimensional parameter-state space. A nonzero polynomial cannot vanish on any such open subset. Therefore the nonvanishing condition is open dense for every $N_0 > 0$. The positive root substitution is an analytic diffeomorphism, so its inverse image is open dense in $\Omega_{N_0}$. No genericity restriction was imposed on the competitor. □

This argument proves the existence of finite rational recovery expressions on the cover. It does not require their full expansion. The upper bound is an application of the cited theory, rather than a new finite-generation theorem.

# 3 Exact positive pairs with the same fourth-order jet

It is convenient to use $h = 1/Z = \beta/(1 + \alpha I)^k$. The coordinates

$$(\alpha, \beta, \gamma, S_0, I_0) \longleftrightarrow (\alpha, \gamma, S_0, y_0, h_0)$$

are analytically invertible on the positive domain, with inverse $\beta = h_0(1 + \alpha y_0)^k$. The rational vector field becomes

$$S' = -Syh, \qquad y' = Syh - \gamma y, \qquad h' = -\frac{k\alpha h y'}{1 + \alpha y}, \qquad \alpha' = \gamma' = 0. \tag{6}$$

Thus every finite jet and its Jacobian is rational in these coordinates for fixed $k$. All Jacobians below use the coordinate order $(\alpha, \gamma, S_0, y_0, h_0)$.

Put $y = y_0$, $v = I'(0)$, $w = I''(0)$, and define

$$\rho = \frac{\alpha}{1 + \alpha y}, \qquad r = v + \gamma y, \qquad F = yw - v^2 + k\rho v y r. \tag{7}$$

Differentiating $r = Syh$ in (6) gives $F = -y^2 r h$. Consequently a positive candidate can be reconstructed as

$$\alpha = \frac{\rho}{1 - \rho y}, \qquad h = -\frac{F}{y^2 r}, \qquad S = -\frac{y r^2}{F}, \qquad \beta = h(1 + \alpha y)^k. \tag{8}$$

The conditions are $0 < \rho < 1/y$, $\gamma > 0$, $r > 0$, $F < 0$, and $S + y < N_0$. Here $\rho$ is a continuous coordinate, unrelated to the integer $q$ in (3).

For $N_0 = 10$, take the true point

$$(\alpha, \gamma, S_0, y_0, h_0) = (1, 1, 2, 1, 1), \qquad \beta = 2^k, \quad R_0 = 7. \tag{9}$$

In every case $I(0) = I'(0) = 1$. The remaining fourth-order data and the true-point determinant are:

| $k$ | $I''(0)$ | $I'''(0)$ | $I^{(4)}(0)$ | $\det DJ_4$ |
|---|---|---|---|---|
| 2 | $-3$ | 4 | 10 | 16 |
| 3 | $-4$ | 13 | $-\frac{97}{2}$ | $\frac{1467}{8}$ |
| $\frac{5}{2}$ | $-\frac{7}{2}$ | $\frac{65}{8}$ | $-\frac{391}{32}$ | $\frac{9495}{128}$ |

**Lemma 5** (Exact certificate)**.** *For each exponent in Theorem 2, there is a distinct positive point at $N_0 = 10$ with the same $J_4$ as* (9)*. Both $5 \times 5$ Jacobians $DJ_4$ are nonsingular. Their fifth output derivatives differ.*

*Certificate and verification.* Let $f_k$ and $g_k$ be the rational polynomials in Appendix A. Define $\rho_*$ to be the unique real root of $f_k$ in the displayed $\rho_*$ interval below. Set $\gamma_* = g_k(\rho_*)$, use $y = v = 1$ and the displayed $w$ in (7), and apply (8) to define the second point. Its removed state is $R_* = 9 - S_*$. The following are outward enclosures of the stored exact rational bounds; each decimal endpoint is itself a rational number.

| $k$ | $\rho_*$ | $\det DJ_4(\theta_*)$ | $I_*^{(5)}(0) - I^{(5)}(0)$ |
|---|---|---|---|
| 2 | $(0.4933991454315, 0.4933991454318)$ | $(-19.70, -19.67)$ | $(-1.01, -0.98)$ |
| 3 | $(0.4654318384721, 0.4654318384724)$ | $(-432.72, -432.69)$ | $(-13.21, -13.18)$ |
| $\frac{5}{2}$ | $(0.4791575947686, 0.4791575947689)$ | $(-129.86, -129.83)$ | $(-5.17, -5.14)$ |

The certificate checker performs these finite exact operations. Sturm's algorithm counts one root in each displayed interval (and in its narrower stored interval) and exactly one root of each $f_k$ in $(0, 1)$. The isolating interval excludes $1/2$. Arithmetic modulo $f_k$ constructs (8); every inverse used is checked by the Euclidean algorithm. Rational interval evaluation gives positive lower bounds for $\alpha_*, \gamma_*, S_*, Z_* = 1/h_*, R_*$. In particular the second point is admissible and distinct from (9). The coordinate enclosures make this second branch explicit:

| $k$ | $\alpha_*$ | $\gamma_*$ | $S_*$ | $Z_*$ | $R_*$ |
|---|---|---|---|---|---|
| 2 | $(0.972, 0.975)$ | $(0.846, 0.849)$ | $(1.567, 1.570)$ | $(0.848, 0.851)$ | $(7.430, 7.433)$ |
| 3 | $(0.869, 0.872)$ | $(0.531, 0.534)$ | $(0.820, 0.823)$ | $(0.535, 0.538)$ | $(8.177, 8.180)$ |
| $\frac{5}{2}$ | $(0.918, 0.921)$ | $(0.653, 0.656)$ | $(1.085, 1.088)$ | $(0.655, 0.658)$ | $(7.912, 7.915)$ |

Here $I_* = 1$ in all three rows, while the true coordinates $(\alpha, \gamma, S, Z, R)$ are $(1, 1, 2, 1, 7)$.

Taylor integration of (6) through order five, using ordinary rational coefficients and factorial conversion, gives zero remainders modulo $f_k$ for all five components of $J_4(\theta_*) - J_4(\theta)$. Differentiating the same recurrence with respect to the five initial coordinates gives both Jacobian matrices. The true determinants are the displayed nonzero rationals. The alternate determinants and the fifth-derivative differences are coprime to $f_k$; their interval enclosures also exclude zero. These computations establish the stated identities and inequalities at the selected real roots. They do not require trusting a numerical root finder or a computed Gröbner basis.

The data are in `producer/output/certificate.json`. Two separately written standard-library checkers use different ODE coordinates and determinant calculations and agree on the polynomial remainders. The exact reduced polynomials and interval bounds are included with the article, so the finite calculation can be repeated. From the packet directory run `python3 -B verify.py`; it replays `independent/check_source_certificate.py` and `producer/verify_certificate.py` with exact arithmetic. □

One route to the candidates is to differentiate $Z = -y^2 r/F$, noting $D\rho = -\rho^2 v$. This gives

$$H = yr\,DF - F\{2vr + y(w + \gamma v) - k\rho v y r\} = 0. \tag{10}$$

For a fourth-order candidate, $H = DH = 0$. The coefficient of $I'''$ in $H$, and of $I^{(4)}$ in $DH$, is $y^2 r > 0$. For the data in (9), elimination of these two equations produces a polynomial in $\rho$ containing the known true-point factor $2\rho - 1$. Removing that factor leaves the quartic $f_k$, which therefore does not include the true point $\rho = 1/2$. The certificate proof above checks the remaining candidate directly in the original ODE. It does not use a complete count of all positive candidates.

# 4 From regular pairs to least orders

**Lemma 6.** *Let $\Phi : U \to \mathbb{R}^d$ be a continuously differentiable map on an open set $U \subset \mathbb{R}^d$. Suppose distinct points $a, b \in U$ have the same image and both $D\Phi(a)$ and $D\Phi(b)$ are nonsingular. Then a nonempty open subset of $U$ consists of points with a distinct competitor under $\Phi$.*

*Proof.* Take disjoint neighborhoods of $a$ and $b$ on which $\Phi$ is a diffeomorphism onto open sets. Those two images contain the common value $\Phi(a) = \Phi(b)$, so their intersection is nonempty and open. Its inverse

image in the first neighborhood is nonempty and open, and every point there has another inverse image in the second. □

*Proof of Theorem 2.* First take $N_0 = 10$. The nonzero true determinant in Lemma 5 shows that the rational function $\det DJ_4$ is not identically zero. Its nonvanishing set is open dense in the positive coordinate domain. The inverse function theorem gives local identifiability there.

For $m \le 3$, $J_m$ maps a five-dimensional open set to dimension $m+1 \le 4$. The maximum-rank locus is open dense: the jet map is rational in the coordinates of (6), so a nonzero maximal minor has a zero set with empty interior. The constant rank theorem gives a positive-dimensional local fiber at every point of that locus. Such a fiber contains other positive admissible points. Thus no open dense set can be locally identifiable for $m \le 3$, and the least generic local order is four.

Apply Lemma 6 to $\Phi = J_4$ and the two points of Lemma 5. The resulting open set of ambiguous true points prevents any open dense set from satisfying global identifiability at order four. Lower orders cannot distinguish points that already share $J_4$. Proposition 4 supplies the order-five upper bound.

Finally let $c = N_0/10 > 0$. The scaling

$$(S, I, R, Z) \mapsto (cS, cI, cR, cZ), \qquad (\alpha, \beta, \gamma) \mapsto (\alpha/c, \beta/c, \gamma) \tag{11}$$

preserves (1), multiplies every output derivative by $c$, and is an analytic diffeomorphism between the two positive parameter domains. It takes the pair above into $\Omega_{N_0}$, preserving jet equality and nonsingularity at both points. The same local and global lower-bound arguments therefore apply for every $N_0 > 0$. □

The conclusion at order four is an open set of ambiguity. It is not a claim that every positive point is ambiguous or that every fiber has exactly two points.

# 5 Additional certificates and scope

A separate interval calculation uses the polynomial coordinates

$$\rho = \frac{\alpha}{1+\alpha y}, \quad b = \frac{\beta}{(1+\alpha y)^k}, \quad s = Sb, \quad g = \gamma,$$

with equations

$$\begin{aligned} y' &= y(s-g), \quad & s' &= -syb - k\rho s y', \\ b' &= -k\rho b y', \quad & \rho' &= -\rho^2 y', \qquad & g' &= 0. \end{aligned} \tag{12}$$

After fixing $y, v, w$, one recovers $g = s - v/y$ and $b = -(w/y - v^2/y^2)/(sy) - k\rho v/y$. Thus the remaining two unknowns $(\rho, s)$ map to $(I''', I^{(4)})$. On the positive domain the change between $(\alpha, \gamma, S, y, h)$ and $(y, v, w, \rho, s)$ is invertible, so nonsingularity of this reduced $2 \times 2$ Jacobian is equivalent to nonsingularity of the full $DJ_4$.

The separate supplementary records give interval certificates for $k = 7/2$ and $k = 1 + \sqrt{2}$, including the changed true point $(\alpha, \gamma, S_0, I_0, Z_0, N_0) = (2/3, 1, 2, 1, 1, 10)$. The alternate $(\rho, s)$ centers are approximately $(0.3707239149, 1.5505151892)$ and $(0.3930391625, 1.8392705345)$, respectively. The certified fifth-derivative differences lie in $(-9.752, -9.750)$ and $(-0.954, -0.952)$. These decimal intervals are outward rational enclosures, not numerical evidence by themselves.

Let $\mathcal{F}(\rho, s)$ be the two residuals between the candidate's third and fourth derivatives and those of the true point. For a rational box $B$ and its center $a$, the checker forms an invertible rational preconditioner $C$ and bounds

$$K(B) = a - C\mathcal{F}(a) + (\mathrm{Id} - C\mathcal{F}'(B))(B - a).$$

It verifies $K(B) \subset \operatorname{int} B$ and $\|\mathrm{Id} - C\mathcal{F}'(B)\|_\infty < 1$. The contraction mapping theorem supplies a unique zero, while interval bounds give positivity, nonsingularity at both branches, and a nonzero fifth-derivative difference. For $1 + \sqrt{2}$ the calculation is uniform over a rational interval whose squared shifted endpoints straddle 2. The irrational exponent is not replaced by one rational approximation. The exact boxes, exponent intervals and rational preconditioners are stored under each exponent key in `independent/stage1-certified-pairs.json` and `independent/extra-pairs.json`; the latter contains the changed initial point. The fields `root_box_p_q`, `k_interval` and `preconditioner` specify these inputs

(the historical $p, q$ there mean $\rho, s$, not the cover integers). Run `python3 -B verify_intervals.py` to reproduce the seven stored inclusions and bounds. These supplementary assertions depend on those separate interval records; the three printed algebraic certificates suffice for Theorem 2.

These additional records corroborate the fourth-order ambiguity mechanism; the main theorem remains restricted to $2, 3, 5/2$. The packet does not classify all exponents or all exceptional points, and it does not supply an expanded global inverse for $J_5$. Unknown $k$, unknown $N_0$, numerical stability and estimating derivatives from noisy observations are outside the result. Nonsingular jet Jacobians give local inverses; they do not bound their condition numbers or guarantee stable estimation.

**AI-use statement.** AI systems assisted with the proof development, separate checks, exact computations, literature comparison and manuscript drafting. The author is responsible for the mathematical claims and their stated scope; AI systems are not authors.

# A Polynomials defining the three main certificates

The alternate point is defined by the unique real root $\rho_*$ of $f_k$ in the displayed interval of Section 3, and by $\gamma_* = g_k(\rho_*)$. The narrower JSON intervals are additional implementation data, not part of this definition. The expressions below are obtained by reducing the recorded $\gamma$ polynomial modulo $f_k$, without changing its value at a root. For $k = 2$,

$$f_k(X) = 1152X^4 - 14112X^3 + 59952X^2 - 119970X + 46225,$$
$$g_k(X) = \frac{12672X^3 - 31392X^2 - 857568X + 1565845}{1340525}.$$

For $k = 3$,

$$f_k(X) = 44000X^4 - 352300X^3 + 1015240X^2 - 1326500X + 430923,$$
$$g_k(X) = \frac{68200000X^3 - 387643000X^2 - 10844350X + 544397116}{867447999}.$$

For $k = \frac{5}{2}$,

$$f_k(X) = 38556X^4 - 364248X^3 + 1223067X^2 - 1885095X + 660490,$$
$$g_k(X) = \frac{3144665916X^3 - 17421405048X^2 - 29056911273X + 71805073975}{82898099900}.$$

# B A proof of cover observability for the three exponents

We prove Lemma 3. This is the rational argument of [4, Sections 2–5], with a direct coefficient-recovery step to make the observation-field assertion explicit.

**Rational constants.** The derivation $E = (z^p/y)D$ has the same constants and satisfies

$$E = -\beta S\partial_S + \left(\frac{\alpha\beta}{q}Sz^{1-q} - \frac{\alpha\gamma}{q}z^{p-q+1}\right)\partial_z.$$

For a nonzero constant $f \in K(z)(S)$, expand $f = \sum_{n\geq m} f_n(z)S^n$ in $K(z)((S))$, with $f_m \neq 0$. The derivation preserves $S$-order, so its lowest coefficient gives $-\beta m f_m - (\alpha\gamma/q)z^{p-q+1}f_m' = 0$. If $m \neq 0$, then $f_m'/f_m$ would have a pole of order $p - q + 1 \geq 2$ at zero. A rational logarithmic derivative has at most a simple pole there. Hence $m = 0$ and $f_0 \in K$. If $f - f_0$ were nonzero, the same argument would contradict its positive $S$-order. Thus $\ker D = K$. This verifies the rational first-integral hypothesis without a numerical test.

**The minimal output relation.** Put $u = 1 + \alpha y$, $v = Dy$, $w = D^2 y$ and

$$A = yuw - \{1 - (k-1)\alpha y\}v^2 + k\alpha\gamma y^2 v.$$

Eliminating $S$ from $v + \gamma y = \beta S y / z^p$ and $DS = -(v + \gamma y)$, by differentiating the former identity, gives

$$z^{p-q} A = -\beta y^2 (v + \gamma y), \qquad Q = u^{p-q} A^q - (-\beta)^q y^{2q} (v + \gamma y)^q = 0. \tag{13}$$

Here $K(S, z) = K(v, z)$, with $v, z$ independent; thus $y, v$ are independent. Over $F = K(y, v)$, $T^q - u$ is Eisenstein at $u$ in $K(v)[u][T]$ (or linear for $q = 1$). The expression for $w$ in (13) is an element of $F$ plus a nonzero element of $F$ times $z^{-(p-q)}$. Since $\gcd(p - q, q) = 1$, its $q$ conjugates are distinct. Therefore $Q$ has minimal degree $q$ in $w$ and is irreducible over $F$. Its leading coefficient in $w$ is $y^q u^p$; the only possible content factors in $K[y, v][w]$ are $y, u$. But $Q(0, v, w) = (-1)^q v^{2q}$ and $Q(-1/\alpha, v, w) = -(-\beta)^q \alpha^{-2q} (v - \gamma/\alpha)^q$ are nonzero. Hence $Q$ is primitive and irreducible in $K[y, v, w]$. The kernel of evaluation $K[y, v, w] \to L$ is exactly $(Q)$: division over $F$ and primitivity give this assertion. The monomial coefficient $[y^q w^q]Q$ is one.

**Coefficients are in the observation field.** Let $m_0 = y^q w^q, m_1, \ldots, m_n$ be the monomial support of $Q$ over $K$, evaluated in $L$, and write $Q = m_0 + \sum_{j=1}^{n} c_j m_j$. A $K$-linear relation among these evaluations lies in $(Q)$. Its total degree is at most $\deg Q$, so it is a scalar multiple of $Q$. In particular $m_1, \ldots, m_n$ are independent over $K$. Since $\ker D = K$, the characteristic-zero Wronskian criterion gives $W = \det(D^i m_j)_{0 \le i < n,\, 1 \le j \le n} \neq 0$ in $L$. For clarity, the criterion follows by induction: divide by a nonzero first element and use $W(f_1, \ldots, f_n) = f_1^n W(1, f_2/f_1, \ldots, f_n/f_1)$; expansion along the first column reduces to the derivatives of the ratios. A constant relation among those derivatives integrates to a constant relation among the original elements. The converse is obtained by differentiating a constant relation.

Differentiate $Q = 0$ for $i = 0, \ldots, n - 1$. Since $Dc_j = 0$, the invertible system $\sum_j (D^i m_j) c_j = -D^i m_0$ expresses each $c_j$ as a rational function of output derivatives. Thus every coefficient of $Q$ belongs to $\mathcal{O}$. This is a field identity, not merely equality of identifiable values on positive real solutions. Although this construction uses derivatives of possibly high order, (4) subsequently places the same quantities in $F_5$; there is no assumed fifth-order inverse in this argument.

**Parameter and state recovery.** For $q = 1$ and $p = 2, 3$, expansion of (13) gives

$$[y^2 w]Q = p\alpha, \quad [y^3 v]Q = p(p-1)\alpha^2\gamma, \quad [y^2 v]Q = \beta + p\alpha\gamma.$$

These recover $\alpha, \gamma, \beta$ successively in $\mathcal{O}$. For $(p, q) = (5, 2)$, the needed coefficients are

$$[y^3 w^2]Q = 5\alpha, \quad [y^3 vw]Q = 5\alpha\gamma, \quad [y^6]Q = -\beta^2\gamma^2.$$

They recover $\alpha, \gamma, \beta^2$ in $\mathcal{O}$. All divisions are by nonzero elements of the rational function field. Finally (13) and $v + \gamma y = \beta S y / z^p$ imply

$$S = -\frac{yu(v + \gamma y)^2}{A} \in \mathcal{O}. \tag{14}$$

The denominator is not identically zero; along a positive solution $z^{p-q} A = -\beta^2 S y^3 / z^p \neq 0$. Also $y \in \mathcal{O}$ by definition. This completes the proof of the lemma.

# References


[1] B. Pant, O. Saucedo and G. Pogudin, "Uncovering identifiability of epidemiological models: basic reproduction number and complementary data streams," medRxiv, version 2 (25 January 2026), doi:10.64898/2026.01.16.26344284.

[2] B. Pant, O. Saucedo and G. Pogudin, Supplementary material to [1], version 2, Section S9, `https://www.medrxiv.org/content/medrxiv/early/2026/01/25/2026.01.16.26344284/DC1/embed/media-1.pdf`.

[3] A. Demin, G. Pogudin and C. Rackauckas, "Observable functions of rational ODE models and how to find them," arXiv:2609.06134v3 (18 September 2026). The numbering used here is that version's.

[4] T. Takimoto, "Global identifiability of a behavioral SIR model for every fixed real saturation exponent $k \geq 2$," article version 3.0 (27 October 2026), publication display edition (10 November 2026), Lemma 3, Lemma 4, Propositions 5–6 and Section 5, equations (14)–(16). The display edition is included in the `companion/` directory of the accompanying packet. The TeX source has SHA-256 `6689d03fc3aaeeb9741b60c4169c831e5ecb3b20eb1884a36baa09985d940d3b`; the PDF has SHA-256 `ab6e2378648e9da9972c19ec79ed9a32c9503bcf8b50f27020cf9d63011b4e14`.

[5] A. Ovchinnikov, G. Pogudin and P. Thompson, "Parameter identifiability and input–output equations," *Appl. Algebra Engrg. Comm. Comput.* **34**(2), 165–182 (2023), doi:10.1007/s00200-021-00486-8; arXiv:2007.14787v2. The theorem numbering used here is that of the cited arXiv version.

[6] H. Hong, A. Ovchinnikov, G. Pogudin and C. Yap, "Global identifiability of differential models," *Comm. Pure Appl. Math.* **73**(9), 1831–1879 (2020), doi:10.1002/cpa.21921; arXiv:1801.08112v5.